

Data Quality Report

South Indian Tourism Analytics & Economic Intelligence

1. Dataset Overview

The final cleaned tourism dataset contains **450** rows and **8** columns. It covers **2005–2022** and contains yearly tourism data for **5** states: Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, Telangana.

Attribute	Details
Total Rows	450
Total Columns	8
Years Covered	2005–2022
Data Frequency	Yearly
States	5
Revenue Available	Yes
Revenue Unit(s)	Crores, Lakhs, Millions

2. Dataset Columns

Column	Description
Year	Year of tourism data
State	State in South India
City	Tourism destination/city
Domestic Visitors	Number of domestic visitors
Foreign Visitors	Number of foreign visitors
Total Visitors	Total visitor count
Revenue	Tourism revenue
Revenue Unit	Unit in which revenue is recorded

3. State and City Standardization

State and city fields were reviewed for formatting inconsistencies. State unique counts were 5 before trimming and 5 after trimming. City unique counts were 27 using the raw strings, 27 after trimming spaces, and 25 after trimming and case-insensitive comparison. The cleaning process removes leading/trailing spaces and standardizes city naming where required.

4. Revenue Normalization

Revenue is present in the dataset. The recorded revenue unit(s) are: **Crores, Lakhs, Millions**. For analysis, revenue should be treated consistently using the recorded unit. Revenue-per-visitor calculations in the Power BI dashboard convert Crores to rupees where required.

5. Missing Value Analysis

The final cleaned CSV was checked for missing values in every column. The table below reports both the count and percentage of missing values.

Column	Missing Values	Missing %
Year	0	0.00%
State	0	0.00%
City	0	0.00%
Domestic Visitors	2	0.44%
Foreign Visitors	4	0.89%
Total Visitors	0	0.00%
Revenue	2	0.44%
Revenue Unit	0	0.00%

Missing tourism data can affect trend calculations, city comparisons, visitor ratios, and revenue-efficiency indicators. Any imputation or removal should be documented consistently with the final cleaned dataset.

6. Data Integrity Checks

Check	Result
Negative Domestic Visitor values	1
Negative Foreign Visitor values	0
Negative Total Visitor values	0
Domestic + Foreign $\neq$ Total mismatches	3
Years covered	2005–2022
Number of states	5

The integrity checks identify invalid negative visitor counts and arithmetic inconsistencies between the domestic, foreign, and total visitor fields. Any remaining anomalies should be considered when interpreting analytical results.

7. Cleaning Actions Performed

- Reviewed the structure, years, states, cities, visitor measures, and revenue fields.
- Checked state and city naming consistency and removed leading/trailing spaces where applicable.
- Checked missing values across all columns.
- Checked visitor counts for negative values.
- Checked whether Domestic Visitors + Foreign Visitors equals Total Visitors.
- Reviewed revenue units and maintained a consistent unit for analysis.
- Verified that the dataset covers yearly observations from 2005 to 2022.

8. Business Risks of Poor Data Quality

Poor-quality tourism data can lead to incorrect estimates of visitor demand, tourism growth, and economic value. Missing or inconsistent records can distort city rankings, domestic-versus-foreign comparisons, revenue-per-visitor calculations, and policy decisions. Reliable and standardized data is therefore important for tourism planning, infrastructure investment, destination marketing, and monitoring tourism recovery or decline.

9. Conclusion

The final cleaned dataset provides a structured basis for SQL analysis and Power BI reporting. Data-quality checks covered dataset structure, naming consistency, revenue units, missing values, negative values, and visitor-total consistency. The identified anomalies should be considered when interpreting tourism trends and economic indicators.